

ARNOB RIZWAN AHMAD

Software Engineering Graduate • AI / ML • Full-Stack • Data & ERP • Cloud

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PROFILE

Software Engineering graduate of Universiti Teknologi Malaysia (July 2026, First Class — CGPA 3.97), currently an Agentic AI Engineer at BLOOMTECH and an ERP Developer at Hirdaramani Bangladesh, where he builds ERP modules and tunes SQL against large operational datasets. Has shipped production software for clients in four countries — ML Engineer for a US news-intelligence platform, application developer for a Luxembourg logistics startup, and founding backend engineer at an AI storage startup. Strong across Python and SQL, LLM/GenAI agents and evaluation pipelines, PyTorch, REST APIs and FastAPI, with certified Microsoft Azure AI and AWS cloud experience, CI/CD automation, and AMD ROCm/HIP GPU work across 93 public repositories. Works directly with international stakeholders to turn business requirements into shipped, maintainable software. Seeking AI/ML, backend or data engineering roles where model optimization, large-scale data handling and LLM-powered automation drive real-world impact.

EDUCATION

Universiti Teknologi Malaysia (UTM) — Johor Bahru

Oct 2022 – July 2026

Bachelor of Software Engineering (Computer Science) — Graduated July 2026 · CGPA 3.97 (First Class Honours)

PROFESSIONAL EXPERIENCE

Agentic AI Engineer — BLOOMTECH, Inc. — Remote (Part-time)

Aug 2026 – Present

Agentic AI systems and LLM-driven tooling

- Building agentic AI workflows in Python — multi-step LLM planning, tool and API integration, and structured output handling.
- Designing prompt and evaluation loops to measure and improve agent reliability across task categories.

Enterprise Resource Planning Developer — Hirdaramani Bangladesh — Hybrid (Full-time)

Jul 2026 – Present

ERP systems for a multinational apparel manufacturing group

- Developing and maintaining ERP modules supporting production, inventory, and order-management workflows across factory operations.
- Writing and tuning SQL against large operational datasets, and building reports and data integrations between ERP and internal systems.
- Working on-site and remote with operations and IT stakeholders to translate process requirements into shipped features.

Software Engineer (Freelance Contributor) — MOMS Mobile Oil Change — USA (Remote)

Apr 2026 – Jul 2026

Production Next.js website for a US mobile oil change & fleet maintenance business — live at momsoilchange.com

- Redesigned the site header and navigation — active-route highlighting, hover dropdowns, prominent phone CTA — and rebuilt three product pages.
- Led an SEO overhaul: dynamic sitemap, 301 redirects for legacy city URLs, and consolidation onto /locations/[city] dynamic routes.
- Ran a site-wide security, dependency, and code-quality audit and repaired production-build breakages.

Application Developer — Moien Delivery — Luxembourg (Remote)

Mar 2026 – Jul 2026

Hybrid logistics platform serving customers, restaurants, and drivers

- Built the Angular web front-end and a cross-platform React Native mobile app on a Nuxt.js backend, delivering features for three distinct user roles.
- Implemented real-time order tracking, driver dispatch, and delivery workflow automation over REST APIs and WebSocket event streams.
- Owned features from spec through QA to production deployment on weekly release cycles with international stakeholders.

Artificial Intelligence Intern — Syntecxhub — Remote (Contract)

Jan 2026 – Feb 2026

Applied AI/ML support for the product engineering team

- Supported applied AI/ML work in Python — dataset preparation and cleaning, model experimentation, and evaluation of results.
- Prototyped LLM-based features and documented findings for the engineering team.

Machine Learning Engineer (Contract via Mercor) — AI News HUD — USA (Remote)

Oct 2025 – Mar 2026

US-based AI-powered news intelligence platform

- Built and evaluated production PyTorch models for real-time news classification, summarization, and personalized recommendations.
- Designed evaluation pipelines and ran iterative experiments to improve model accuracy and inference latency across live traffic.
- Took ML components from prototype to deployed feature in the platform's core intelligence layer.

Founding Backend Engineer — Framic — Remote

Aug 2025 – Jul 2026

AI-powered storage application

- First backend hire — designed backend services from scratch and owned architecture decisions and core API design.
- Led development of AI search, real-time file indexing, and scalable systems for gallery and drive functionality.
- Worked directly with founders to ship features rapidly while keeping the system maintainable as the team scaled.

AI Trainer — Braintrust — Remote (Freelance)

Apr 2025 – Jul 2025

Human feedback and evaluation data for large language models

- Evaluated and ranked large language model responses against rubric criteria on technical and coding prompts.
- Authored corrected reference answers and structured feedback used as model training and evaluation data.

Software Engineering Resident — Headstarter — Remote (US)

Sep 2024 – Mar 2025

- Completed a competitive 6-month engineering residency, shipping 14+ ML and full-stack projects in Python and TypeScript.
- Designed and trained 5 neural networks applying LLM chaining, hyperparameter tuning, and model fine-tuning techniques.
- Built LLM-powered agents and automation workflows on AWS Bedrock/OpenAI APIs and optimized a Docker-based CI/CD pipeline on GitHub Actions, significantly reducing image size and build times.

SELECTED PROJECTS

Autonomous ML Agent — LLM-Planned AutoML Pipeline (GitHub)

- Built an AutoML system where an LLM planner proposes modeling strategies under a time budget, Optuna tunes XGBoost/LightGBM/scikit-learn models, ensembles are built, and SHAP explains the winner.
- Shipped with model cards, a FastAPI inference service, 5 pytest suites, and CI/CD publishing to Docker Hub.

Self-Improving Prompt Optimization API — LLM Evals Infrastructure (GitHub)

- Built CI/CD for prompts: a semver-versioned prompt registry where each improvement cycle runs failure analysis, generates candidates, A/B-evaluates them with a blinded LLM judge, and auto-promotes winners per YAML rules.
- Multi-dimensional scoring across OpenAI and Anthropic providers with FastAPI, Postgres, and Redis.

CrisisCohort — Multi-Agent AI Climate Resilience Platform (GitHub)

- Architected a climate-resilience platform for smallholder farmers with 10+ specialized AWS Bedrock agents (water optimization, climate migration, grant matching, disease diagnostics) across 15,000+ lines of Lambda.
- Designed an AWS architecture spanning Lambda, Bedrock, DynamoDB, CloudFront, API Gateway, and Cognito, plus an on-chain water-rights trading contract in Solidity.

Terminal Coding Agent — Sandboxed CLI Agent (GitHub)

- Built a pip-installable CLI coding agent that plans edits via OpenAI/Anthropic, applies them through a diff manager, and executes inside E2B Firecracker microVMs.
- Added guardrails scanning for six secret types (AWS, GitHub, Slack keys) and 5 pytest suites.

School Care App — 40-School Deployment — Sylhet, Bangladesh

Client work

- Delivered a full-stack school management system adopted by 40 schools across Sylhet; in closed testing ahead of Google Play launch.
- Built a React admin dashboard and a React Native app for parents and teachers, with Cloud Functions fanning out role-targeted FCM push notifications by school, role, and class.

ShopLocalr — Local Commerce Mobile App — Malaysia

Client work

- Shipped a 33,000-line Flutter local-business discovery app with customer, business, and admin roles — QR deal redemption with checkout PINs, Gemini-backed recommendations, and offline-first Hive caching on a Firebase backend.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C#, C++, Dart

ML / AI: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna, SHAP, AWS Bedrock, Gemini, multi-agent systems, LLM evaluation & judges, RAG (FAISS, pgvector), local LLMs (Ollama, Llama 3), prompt engineering, A/B testing, model optimization & profiling

GPU / HPC: AMD ROCm, HIP, CUDA→HIP porting, rocBLAS/MIOpen/rocFFT, rocprofv3 profiling, mixed-precision (FP16/BF16) GEMM, kernel tuning

Backend: FastAPI, Django + DRF, Node.js, Express, Nuxt.js, ASP.NET Core, Celery, REST APIs, WebSockets, Socket.IO

Frontend & Mobile: React, Next.js, Angular, React Native (Expo), Flutter (Riverpod), Tailwind CSS

Cloud & DevOps: Microsoft Azure (Azure OpenAI Service, NLP, Document Intelligence), AWS (Lambda, Bedrock, DynamoDB, API Gateway, Cognito, CloudFront), GCP (Cloud Run), Linux, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Firebase, Redis

Data & ERP: ERP modules (production, inventory, order management), SQL tuning on large operational datasets, reporting and data integrations between ERP and internal systems, relational data modelling

Databases: PostgreSQL, MySQL, SQL Server, DynamoDB, Firestore, Neo4j, pgvector

Automation: Zapier, n8n, Make

CERTIFICATIONS

- **AMD — ROCm Certified Associate** (Aug 2026)
- **AWS** — Building Language Models on AWS; Customizing & Evaluating LLMs Using Amazon SageMaker JumpStart; Amazon SageMaker JumpStart Foundations; Foundations of Prompt Engineering (Mar 2025)
- **IBM** — Full Stack Software Developer Professional Certificate (Aug 2024)
- **Microsoft Azure AI** — Azure OpenAI Service, Natural Language Processing, Document Intelligence (Jun 2024)

LEADERSHIP

Director — ISS Central UTM

Jan 2025 – Sep 2025

Head of Department (Academic) — ISS Bangladesh, UTM

Oct 2024 – Sep 2025

ACHIEVEMENTS

- Earned the **AMD ROCm Certified Associate** certification with 100% on all five course quizzes and the 30-question final exam, first attempt.
- Selected as a Headstarter Software Engineering Resident — competitive 6-month US-based engineering program; shipped 14+ ML and full-stack projects.
- Contributor to two live production websites in the US — momsoilchange.com and photonixresearch.com.
- Shipped software used by real users in 4 countries (US, Luxembourg, Bangladesh, Malaysia) across 93 public GitHub repositories.